



Unit 4 · Outcome 3 · Scientific Investigation

U.O3 · KK10 · Science communication conventions — Digital Vector EnviroSci

KK10

VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary with worked good-vs-bad answers, exam traps, command words and practice.



The shared language of science

01 The shared language of science A scientific investigation is only finished when someone else can read it, trust it, and repeat it. Conventions of science communication are the agreed rules that let any scientist — in Mildura or Munich — read your poster and understand it exactly the same way. Get them right and your meaning travels cleanly. Get them wrong and a brilliant investigation can read as careless. CO_2 mg/L $\text{W}\cdot\text{m}^{-2}$ $\mu\text{S}/\text{cm}$ Drag to orbit · the same conventions every scientist agrees to use 3D not available — the orbiting tokens are decorative; ever

Precise terminology beats everyday words

02 Precise terminology beats everyday words Everyday language is vague on purpose — it leaves room for interpretation. Science wants the opposite: one precise term that names exactly what you mean. Swapping casual phrasing for the correct technical term is the quickest mark-saver on a poster. Translator · everyday → scientific tap a phrase Tap Priya's casual phrases to reveal the precise term an examiner rewards. Pick a phrase above Each casual phrase maps to a specific, testable scientific term. Analogy · ordering coffee “A coffee, thanks” gets you anyt

Choosing & labelling the right representation

03 Choosing & labelling the right representation A representation is any non-prose way of showing information — a table, graph, flowchart, food web or labelled diagram. The convention has two halves: pick the form that fits the data , then label it so it stands alone . Representation matcher pick the best form For each of Priya's data situations, choose the representation a marker would expect. Score: 0 / 5 correct Line graph Continuous independent variable (distance, time, temperature). Plot IV on the x-axis , DV on the y-axis . Use a line of best fit,

Symbols, formulas & subscripts

04 Symbols, formulas & subscripts Chemical symbols and formulas are international shorthand. The subscript tells you how many atoms of the element before it — and the case of every letter carries meaning. CO (carbon monoxide) is not CO_2 (carbon dioxide) and neither is Co (cobalt). Formula Reads as Where it appears in EnviroSci CO_2 1 carbon, 2 oxygen Enhanced greenhouse effect; respiration; combustion CH_4 1 carbon, 4 hydrogen Methane — high warming potential N_2O 2 nitrogen, 1 oxygen Nitrous oxide — agriculture, fertilisers O_2 / O_3 oxygen gas / ozone Disso

Formula	Reads as	Where it appears in EnviroSci
CO_2	1 carbon, 2 oxygen	Enhanced greenhouse effect; respiration; combustion
CH_4	1 carbon, 4 hydrogen	Methane — high warming potential
N_2O	2 nitrogen, 1 oxygen	Nitrous oxide — agriculture, fertilisers
O_2 / O_3	oxygen gas / ozone	Dissolved oxygen; stratospheric ozone
H_2O	2 hydrogen, 1 oxygen	Water — the hydrosphere

Standard abbreviations

05 Standard abbreviations An abbreviation only works if it is standard and defined on first use . Write the term in full the first time with the abbreviation in brackets — “dissolved oxygen (DO)” — then use DO thereafter. Inventing your own (“H2Oxy”) defeats the purpose. Tap each card to flip and reveal what the abbreviation stands for. Exam habit If you use an abbreviation in a short-answer response, define it once. Markers will not assume EC means electrical conductivity rather than, say, the European Commission. One bracketed definition protects every

Units of measurement & the SI

06 Units of measurement & the SI The International System of Units (SI) is the global standard. A handful of base units are defined first; everything else is a derived unit built by multiplying or dividing them. Reporting in SI is what makes your number comparable to anyone else's. Base quantity SI base unit length metre (m) mass kilogram (kg) time second (s) temperature kelvin (K) amount of substance mole (mol) Derived quantity Built from area $m^2 = m \times m$ volume m^3 , or litre (L) concentration $mg/L = \text{mass} \div \text{volume}$ density $kg/m^3 = \text{mass} \div \text{volume}$ radiative

Base quantity	SI base unit
length	metre (m)
mass	kilogram (kg)
time	second (s)
temperature	kelvin (K)
amount of substance	mole (mol)

Derived quantity	Built from
area	$m^2 = m \times m$
volume	m^3 , or litre (L)
concentration	$mg/L = \text{mass} \div \text{volume}$
density	$kg/m^3 = \text{mass} \div \text{volume}$
radiative forcing	$W/m^2 = \text{power} \div \text{area}$

SI prefixes & scale

07 SI prefixes & scale Environmental data spans enormous ranges — micrograms of a pollutant up to gigatonnes of carbon. Prefixes rescale a unit by powers of ten so the number stays readable. The quantity never changes — only how you write it. Prefix ladder step the scale 3D not available — the ladder controls and live readout below work on their own. ▼ smaller prefix ▲ larger prefix — A fixed mass of carbon, written with different prefixes — same quantity, different size of number. Prefix Symbol Factor Example in EnviroSci giga G 10^9 Gt CO₂ — global emis

Prefix	Symbol	Factor	Example in EnviroSci
giga	G	10^9	Gt CO ₂ — global emissions
mega	M	10^6	Mha — area of forest cleared
kilo	k	10^3	kW, kJ — energy
milli	m	10^{-3}	mg/L — dissolved oxygen
micro	μ	10^{-6}	μS/cm — conductivity; μg/L — trace pollutant

Notation rules examiners notice

08 Notation rules examiners notice A few small typographic rules separate a polished poster from a rushed one. They feel fussy — but each one removes a possible misreading. Space the unit One space between the number and the unit: 25 °C , 8.2 mg/L — not 25°C or 8.2mg/L . (Exceptions: % and pH carry no spaced unit.) Don't pluralise symbols 5 kg , never 5 kgs . Unit symbols are invariant — they never take an “s” and never take a full stop (kg , not kg.). Case is not optional kg , L , K . Units named after a person take a capital symbol but lowercase name

Significant figures & scientific notation

09 Significant figures & scientific notation Your numbers must be honest about precision . Significant figures show how finely you actually measured; scientific notation keeps very large or very small numbers compact and unambiguous. Counting significant figures Non-zero digits always count. Zeros between them count. Leading zeros never count (0.0048 → 2 s.f.). Trailing zeros count only with a decimal point (8.20 → 3 s.f.). Reporting a result A calculated answer can be no more precise than your least precise measurement. If your probe reads to 2 s.f., do

Command-word decoder

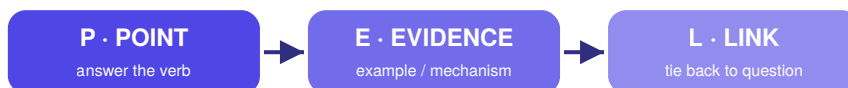
◆ COMMAND WORDS

10 Command-word decoder Convention questions often hinge on the verb. Convert wants working and a unit; justify wants a reason. Match your answer's shape to the command word. Tap a command word to see what the examiner expects.

P·E·L writing trainer

P·E·L WRITING

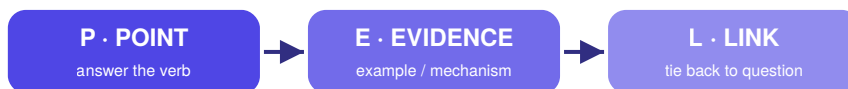
12 P·E·L writing trainer Structure a convention answer as Point · Example · Link : name the convention, show it applied (often a corrected version), then link to why it matters for communication. Type below — the meters react to what you write. Practice prompt On her poster Priya writes: “temp was 25°C and DO was 8mg/l” . Identify two conventions she has breached and rewrite the phrase correctly , explaining why each correction aids communication. (4 marks) Point Name the actual convention Example Show the corrected form Link Why it helps the reader Reve



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

Worked answer · weak vs strong

13 Worked answer · weak vs strong Same question, two responses. The difference is entirely in convention discipline. Question (3 marks) Priya measured dissolved oxygen at six distances downstream. Identify the most appropriate graph type and justify your choice, referring to the variables. State how the axes should be labelled. × Weak response “A bar graph because it shows the oxygen at each spot. Put distance and oxygen on it.” Wrong form (distance is continuous, not categorical), no link to variable type, and axis labelling has no quantities or units.



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

Practice questions · 20 marks

★ PRACTICE & SELF-MARK

14 Practice questions · 20 marks VCAA-style. Attempt each in your logbook first, then reveal the weak answer, strong answer and marking guide. Mark yourself honestly — the dial tracks your total. 0 / 20 score: 0 / 20 Reveal each model answer, then tap the marks you'd award yourself.

One-screen cheat sheet

♦ RECAP / CHEAT SHEET

15 One-screen cheat sheet Convention quick-reference Terminology — one precise technical term, never casual phrasing. Representation — line (continuous IV) · bar (categories) · scatter (relationship) · flowchart (process) · food web (feeding). Title + axes with units + legend. Symbols / formulas — case & subscript carry meaning: CO₂ ≠ CO ≠ Co . Define every symbol in a formula. Abbreviations — define in full on first use: dissolved oxygen (DO). SI units — base (m, kg, s, K, mol); derived built from them (mg/L, W/m²). Prefixes — G 10⁹ · M 10⁶ · k 10³ · m

★ Good vs bad worked answers

Study the contrast — the same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer structure — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

Over-precise Priya's probe reads to 0.1 mg/L. She reports the mean DO as 6.47833 mg/L . Claims five-figure precision from a one-decimal instrument — dishonest and mark-losing.

✓ STRONG / MODEL ANSWER

Honest She reports the mean DO as 6.5 mg/L (2 s.f.), matching the probe's precision. Precision reported = precision measured. An examiner reads this as careful science.

✗ WEAK ANSWER

A bar graph because it shows the oxygen at each spot. Put distance and oxygen on it." Wrong form (distance is continuous, not categorical), no link to variable type, and axis labelling has no quantities or units. ~0–1 / 3.

✓ STRONG / MODEL ANSWER

A line graph is most appropriate because distance downstream is a continuous independent variable and dissolved oxygen is a continuous dependent variable, so a line can show the trend. The x-axis should read Distance downstream (m) and the y-axis Dissolved oxygen (mg/L) , each with its unit in brackets." Correct form, justified by variable type, axes labelled with quantity + unit. 3 / 3.